THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 7 & 8: **Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 480 J	W = F × s
2	**(b)** 50 J	½mv²
3	**(d)** 300 J	mgh
4	**(c)** 250 W	P = W/t
5	**(c)** 5	L ÷ h
6	**(b)** 20 Hz	oscillations ÷ time
7	**(c)** 200 m s^{-1}	v = ν λ
8	**(b)** 340 m s^{-1}	v = ν λ
9	**(d)** 85 m	distance = vt/2
10	**(a)** 3000 m	d = vt/2

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	mgh = 5 × 10 × 4 = 200 J (2).	2
12	In the absence of other forces, the sum of KE and PE stays constant (2).	2
13	Maximum change in density from the average (1); larger amplitude gives louder sound (1).	1+1
14	Pitch depends on frequency (shrill/deep); loudness depends on amplitude (1+1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	½mv_1^2 = ½(4m)v_2^2 ⇒ $v_1^2 = 4v_2^2$ (2); $v_1 : v_2 = 2 : 1$ (1).	2+1
16	In a small room reflections return in less than 0.1 s so are not separated (1.5); in a hall the gap is longer, so echoes are heard (1.5).	1½+1½
17	Work doubles (1.5); power is the same (W/t = 2W/2t) (1.5).	1½+1½
	OR: v = 20 m s^{-1}, KE = 200 000 J (2); P = 20 000 W (1).	2+1
18	Ultrasonic pulses sent and reflected (1); d = 1530 × 0.45 (1) = 688.5 m (1).	1+1+1
	OR: Bats emit ultrasonic waves (1); sense the echoes from objects (1); judge the distance and size of obstacles and prey (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Devices that make work easier by changing the magnitude or direction of a force (1). (ii) Load/effort (1). (iii) Fixed: only changes the direction, MA = 1; movable: MA > 1, lifts heavy loads with less effort (3).	1+1+3
	OR: (i) A longer path means a smaller force, but the same work (1). (ii) F′L = mgh ⇒ MA = mg/F′ = L/h (2). (iii) MA = 5, effort = 100 N (2).	1+2+2
20	(i) 340 m (1). (ii) 680 m (1). (iii) Speed increases, so the time decreases (1). (iv) No; 0.03 s is less than 0.1 s (2).	1+1+1+2
	OR: (i) Waves above 20 kHz (1). (ii) Imaging, kidney stone breaking, cleaning, flaw detection, sonar (2). (iii) Echolocation (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 7 & 8: **Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** joule	N m
2	**(a)** four times	KE ∝ v²
3	**(c)** 1	effort = load
4	**(c)** the magnitude or direction of the force	no energy is created
5	**(b)** a stretched bow before release	deformation stores energy
6	**(d)** 1700 m	d = v × t
7	**(a)** compression	particles crowded
8	**(a)** amplitude	larger amplitude is louder
9	**(d)** timbre	overtone pattern
10	**(d)** imaging internal organs	medical imaging

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Rate of doing work (1); watt (1).	1+1
12	W = 75 × 10 × 2 = 1500 J (1); P = 1500/5 = 300 W (1).	1+1
13	Infrasonic (< 20 Hz), audible (20 Hz–20 kHz), ultrasonic (> 20 kHz) (2).	2
14	Reflected sound heard after the original (1); time gap at least 0.1 s (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	KE is due to motion, e.g. a rolling ball (1.5); PE is due to position or shape, e.g. a stretched spring (1.5).	1½+1½
16	Medical imaging; breaking kidney stones; cleaning delicate parts; detecting flaws in metal blocks (1+1+1).	1+1+1
17	L = √(30² + 40²) = 50 cm (1.5); MA = 50/30 = 1.67 (1.5).	1½+1½
	OR: Effort is smaller because it is applied over a longer distance (2); work done (ignoring friction) is the same (1).	2+1
18	Quality of sound (1); overtone patterns differ because of the shape, material and construction (2).	1+2
	OR: Unwanted or harmful sound (1); sleep disturbance, hearing loss (1); sound-proofing, limiting volume (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 200 × 3 = 600 N m (1). (ii) 300 × d = 600 ⇒ d = 2 m (2). (iii) First class (1). (iv) The heavier child's side goes down (1).	1+2+1+1
	OR: (i) Effort × effort arm = load × load arm (1). (ii) Coins in the pans; the heavier pan moved closer to balance (2). (iii) I: scissors; II: wheelbarrow; III: tweezers (2).	1+2+2
20	(i) Single frequency vs mixture; timbre = quality (2). (ii) Different overtone patterns (1). (iii) Frequency ratio 2 : 1 (1). (iv) C. V. Raman (1).	2+1+1+1
	OR: (i) Unwanted or harmful sound (1). (ii) About 60 dB and over 100 dB (1). (iii) Hearing loss, sleep loss, stress (1). (iv) Use of earplugs, sound barriers, limiting volume (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET C

Chapters 7 & 8: **Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 200 J	W = F × s
2	**(a)** 50000 J	½mv²
3	**(d)** 800 J	mgh
4	**(a)** 300 W	P = W/t
5	**(b)** 4	L ÷ h
6	**(b)** 0.25 s	T = 1/ν
7	**(a)** 340 m s⁻¹	v = ν λ
8	**(c)** 2 m	λ = v/ν
9	**(c)** 340 m	distance = vt/2
10	**(d)** 1530 m	d = vt/2

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The ratio of load to effort (2).	2
12	It changes the direction of effort, making pulling down convenient; MA = 1 (2).	2
13	The persistence of sound due to repeated reflections, with gaps less than 0.05 s (2).	2
14	Imaging organs, breaking kidney stones, cleaning parts, flaw detection (any two) (1+1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	A longer path means a smaller slope, hence a larger MA and less effort (3).	3
16	Microphone: sound makes a membrane vibrate, converted to an electrical signal (1.5); speaker: electrical signal makes a cone vibrate and produce sound (1.5).	1½+1½
17	Scissors: fulcrum between (1); wheelbarrow: load between (1); tweezers: effort between (1).	1+1+1
	OR: 15 × 2 = 30 × d (2); d = 1 m (1).	2+1
18	Sound cannot travel in vacuum, so no sound would be heard (3).	3
	OR: ν = 344/3.44 = 100 Hz (1.5); T = 0.01 s (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 20 m s⁻¹ (1). (ii) 2 × 10⁶ J (1). (iii) 2 × 10⁶ = 50 000 d + 10 000 × 10 × d/2 ⇒ d = 20 m (2). (iv) Sand gives a large opposing force and dissipates energy (1).	1+1+2+1
	OR: (i) Mechanical, heat, light, sound, electrical, chemical, nuclear (3). (ii) Water's PE → KE → turbine → electrical (1). (iii) A water-driven mill that uses falling water to turn a grinding stone (1).	3+1+1
20	(i) Eardrum, small bones, cochlea, nerve to the brain (3). (ii) The brain compares arrival times to find the direction (2).	3+2
	OR: (i) Fastest in steel (1). (ii) 4.36 and 3.33 (2). (iii) Particles are more closely packed and interact more strongly in solids (2).	1+2+2

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ANSWER KEY & MARKING SCHEME SET D

Chapters 7 & 8: **Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** watt	1 W = 1 J s^{-1}
2	**(b)** potential energy	deformed shape
3	**(a)** the fulcrum is between effort and load	first class
4	**(d)** 1 kg m^2 s^{-2}	N × m
5	**(b)** 1000 W	kilo = 10^3
6	**(d)** vibrating objects	to and fro motion
7	**(a)** above 20 kHz	above audible range
8	**(c)** frequency	high frequency, high pitch
9	**(b)** 0.1 s	persistence of hearing
10	**(d)** decreases	λ = v/ν

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Fulcrum, load, effort (2).	2
12	Effort × effort arm = load × load arm (2).	2
13	Locating objects by reflected sound (1); bats and dolphins (1).	1+1
14	Loudness level (1); normal conversation about 60 dB, firecrackers over 100 dB (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Heat transfer from hot to cold bodies (1.5); radiation, e.g. the Sun's energy (1.5).	1½+1½
16	Distance = 2.4 m (1); t = 2.4/345 (1) ≈ 0.007 s (1).	1+1+1
17	KE = ½ × 10 000 × 400 = 2 × 10^6 J (2); the sand does negative work and the truck gains PE (1).	2+1
	OR: Real machines lose energy to friction and air resistance as heat and sound, so they slow down; energy cannot be created (3).	3
18	Air: 1 s (1); steel: 0.068 s (1); difference 0.932 s (1).	1+1+1
	OR: Sound makes the eardrum vibrate (1); small bones amplify the vibrations (1); the cochlea converts them into electrical signals sent to the brain (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Energy is conserved; machines only change force/direction (1). (ii) F × 5 = 1000 × 1, F = 200 N (2). (iii) 1000 J (2).	1+2+2
	OR: (i) Positive: pushing a wheelchair; negative: friction on a sliding coin (2). (ii) Energy ∝ mass: 160 : 200 = 4 : 5 (3).	2+3
20	(i) Sonar and ocean temperature study, or soil microbe sounds (3). (ii) Detecting drones from their hum; detecting earthquakes and storms from infrasound (2).	3+2
	OR: (i) 1 s (1). (ii) 0.068 s (1). (iii) 0.932 s (1). (iv) Yes, since the gap exceeds 0.1 s (2).	1+1+1+2

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